WEEK 6

Evaluation in Information Retrieval

Measuring Success: From Precision to User Happiness

Course Roadmap

WEEK 1-2
Intro & Preprocessing

WEEK 3
Inverted Indexes

WEEK 4
Index Construction

WEEK 5
Vector Space Model

WEEK 6
Evaluation Metrics

WEEK 7
Web Crawling

WEEK 8
Link Analysis

WEEK 9-12
Advanced Topics

We are here: Transitioning from *building* search engines to *measuring* their success. How do we know if our system is actually good?

Why Evaluate?

> *"You can't improve what you don't measure."*
>
> — **Peter Drucker**

- **Compare:** Is Algorithm A better than B?
- **Monitor:** Did the update break search?
- **Optimize:** Tune parameters (e.g., BM25).

+15%

PRECISION GAIN

The Cranfield Paradigm

The standard model for laboratory evaluation since the 1960s. It allows for **repeatable** and **comparable** experiments.

1. DOCUMENTS

A fixed, static collection of documents (corpus).

2. QUERIES

A set of information needs (topics) representing user intent.

3. JUDGMENTS

Binary relevance assessments (Relevant / Not Relevant) for each pair.

Key Assumption: Relevance is objective, binary, and static. It does not change based on user context or time.

Test Collections

Document Corpus

D

The fixed set of documents to be searched. Must be static to ensure reproducibility.

EXAMPLE
1 Million News Articles (NYT, AP)

Test Topics

Q

Formal descriptions of information needs. Not just keywords, but intent.

EXAMPLE
Topic 401: "Foreign investment in India"

Relevance Judgments

R

The "Ground Truth". Human assessors determine which docs are relevant.

EXAMPLE
Doc 123 is **Relevant** to Topic 401

Relevance Judgments

The Ground Truth: A human assessment of whether a specific document d satisfies the information need of query q.

Binary Relevance

Simple, unambiguous. Used for standard metrics like Precision & Recall.

1 (Relevant) vs. 0 (Not Relevant)

Graded Relevance

Nuanced. Captures "perfect" vs "okay" answers. Used for NDCG.

Perfect (4) > Excellent (3) > Good (2) > Fair (1) > Bad (0)

Topic Experts (NIST) **Crowdsourcing (MTurk)** **User Clicks (Implicit)**

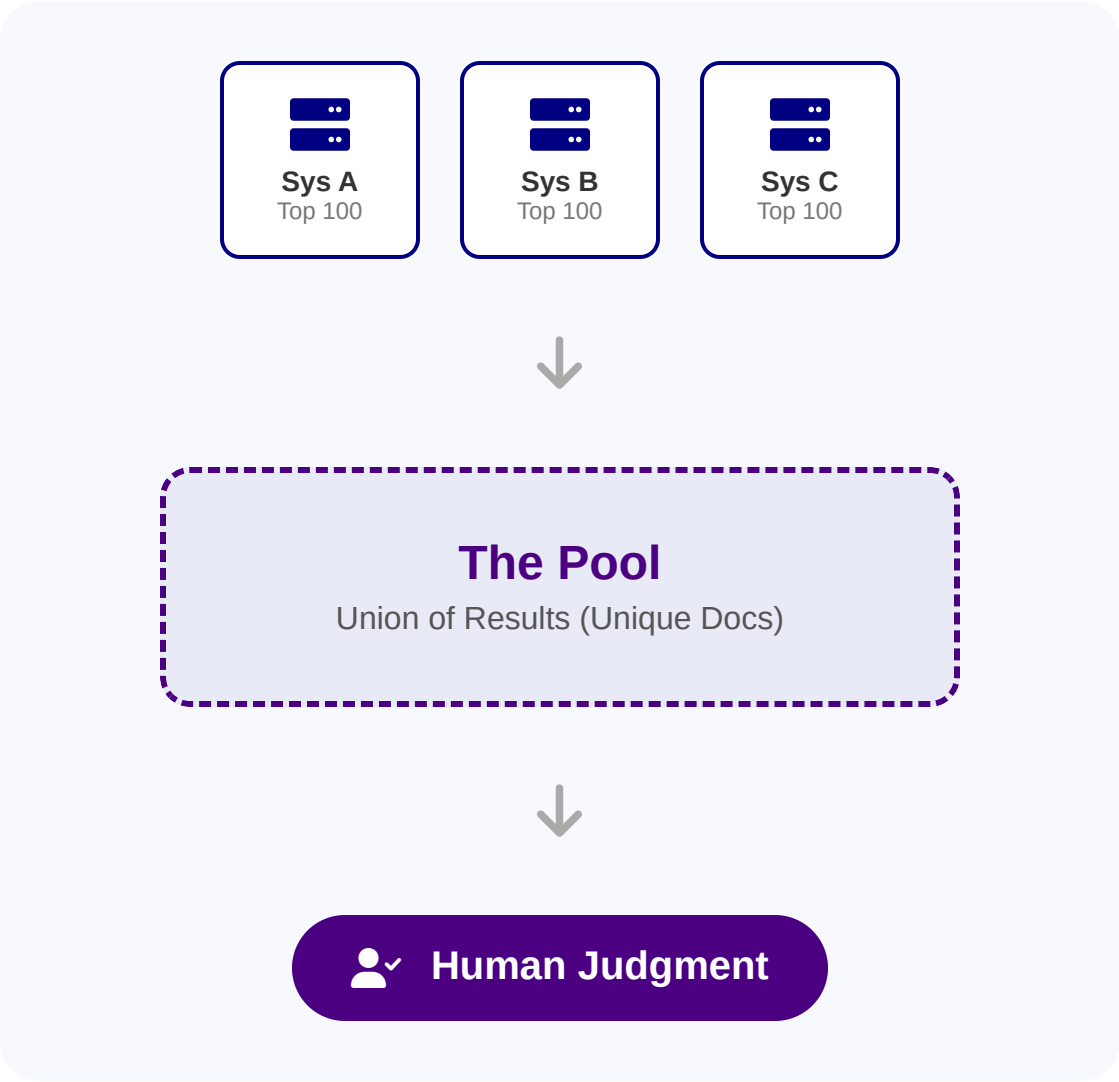
The Pooling Method

THE CHALLENGE

With millions of documents, it is impossible to judge relevance for **every** document-query pair.

1. Run **diverse** retrieval systems (e.g., BM25, Vector, Boolean).
2. Take the **top k** results (e.g., top 100) from each system.
3. **Union** these sets to form a "Pool" (remove duplicates).
4. Human assessors judge **only** this pool.

Assumption: Documents not retrieved in the top k by any system are likely non-relevant.



Binary Evaluation Metrics

The Binary Assumption

The simplest evaluation model assumes a document is either **useful** or **useless**. No in-between.

RELEVANT (1)

NOT RELEVANT (0)

Precision

What fraction of retrieved documents are relevant?

Recall

What fraction of relevant documents were retrieved?

F-Measure

The harmonic mean of Precision and Recall.

The Confusion Matrix

System Output (Retrieval)	Ground Truth (Relevance): Relevant	Ground Truth (Relevance): Not Relevant
Retrieved	**TRUE POSITIVE (TP)** Correctly identified relevant documents.	**FALSE POSITIVE (FP)** "Noise". Irrelevant docs returned to the user.
Not Retrieved	**FALSE NEGATIVE (FN)** "Missed". Relevant docs the system failed to find.	**TRUE NEGATIVE (TN)** Correctly ignored irrelevant docs. (The vast majority).

Precision & Recall

PRECISION

The fraction of **retrieved** documents that are actually **relevant**.

$$P = \mathrm{TP} / (\mathrm{TP} + \mathrm{FP})$$

"How useful is the result list?"

RECALL

The fraction of **relevant** documents that were successfully **retrieved**.

$$R = \mathrm{TP} / (\mathrm{TP} + \mathrm{FN})$$

"Did we miss anything important?"

The Trade-off

The Tug of War: Generally, as you broaden your search to increase **Recall**, you inevitably retrieve more non-relevant documents, lowering **Precision**.

Strict Threshold (High τ)

Only the most confident matches.
High Precision, Low Recall.

Lenient Threshold (Low τ)

Let more documents in.
Low Precision, High Recall.

F-Measure (F1 Score)

A single metric that trades off Precision (P) and Recall (R). It is the **Harmonic Mean** of the two.

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

> ***Why Harmonic?*** *It is a "conservative" average. It is always closer to the minimum of the two numbers, punishing extreme values.*

Why not Arithmetic Mean?

Scenario: A system returns *every* document in the collection.

✓ Recall = 100% (1.0)
✕ Precision $\approx$ 0% (0.01)

Metric	Calculation	Result
Arithmetic Mean	**(1.0 + 0.0) / 2**	**~~50%~~**
Harmonic Mean	**(2 * 1 * 0) / 1**	**0%**

The Harmonic Mean correctly identifies this as a bad system.

Calculating F1-Score

SCENARIO

Total Relevant: **20**

Retrieved: **10**

Relevant Retrieved: **6**

1

Precision (P)

6 / 10

= 0.6

2

Recall (R)

6 / 20

= 0.3

3

F1-Score

2 * P * R / (P + R)

2 * 0.6 * 0.3 / 0.9

0.36 / 0.9

= 0.40

Why Not Accuracy?

Imagine a "Lazy" Search Engine that returns **zero results** for every query.

99.999% Accuracy

Is this a good system? **No.**

THE SCENARIO

Collection Size:	**1,000,000 docs**
Relevant Docs:	**10 (The "Needle")**
Non-Relevant:	**999,990 (Haystack)**
System Action:	**Returns Nothing**

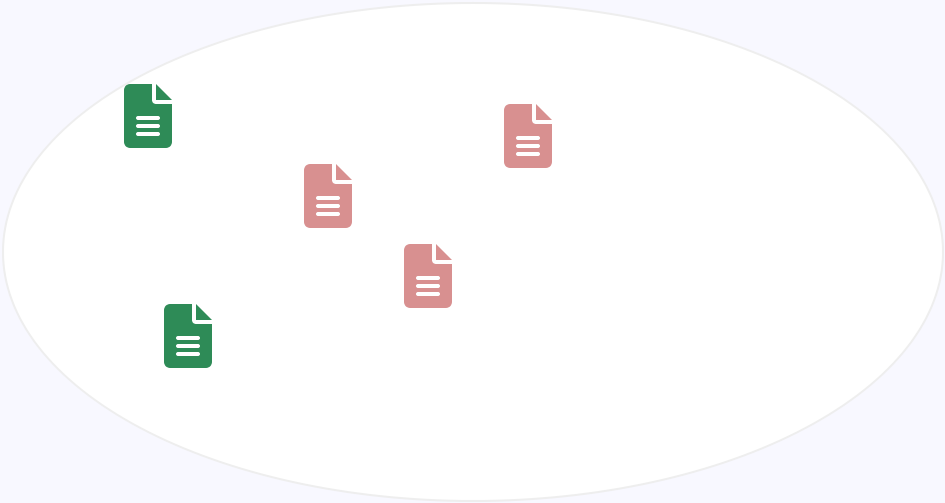
THE CALCULATION

True Positives (TP):	**0**
True Negatives (TN):	**999,990**

```
Acc = (TP + TN) / Total
Acc = (0 + 999,990) / 1,000,000
Acc = 0.99999
```

Evaluating Ranked Retrieval

SET-BASED (BOOLEAN)



"Order doesn't matter."
A relevant document is counted equally whether it's 1st or 100th.

RANKED RETRIEVAL

1. Relevant Doc ✓
2. Relevant Doc ✓
3. Non-Relevant ✗

"Top is critical."
Users scan from the top. A system with relevant docs at #1 & #2 is **better**.

Insight: We need metrics that penalize relevant documents appearing lower in the list.

Precision at K (P@k)

Definition: The proportion of relevant documents within the top k retrieved results. It ignores everything below rank k.

$$P@k = \frac{\text{# Relevant in top } k}{k}$$

- **User-Centric:** Matches real behavior. Users rarely look past the first page (e.g., $k=10$).
- **Interpretability:** "60% of the top 10 results are useful" is easy to explain.
- **Rank Insensitive:** It treats Rank 1 and Rank k equally. (A relevant doc at #10 is "worth" the same as at #1).
- **Unstable:** Highly sensitive to the choice of k.

EXAMPLE: P@5

1	Doc A (Rel)
2	Doc B (Non)
3	Doc C (Rel)
4	Doc D (Non)
5	Doc E (Rel)
6	Doc F (Non)
7	Doc G (Rel)
8	Doc H (Non)

P@5
3 / 5 = 0.6

Note: The relevant doc at Rank 7 is ignored.

Calculating Precision at K

1. Doc A (Relevant)
2. Doc B (Non-Rel)
3. Doc C (Relevant)
4. Doc D (Relevant)
5. Doc E (Non-Rel)

k = 5

6. Doc F (Non-Rel)
7. Doc G (Non-Rel)
8. Doc H (Relevant)
9. Doc I (Non-Rel)
10. Doc J (Non-Rel)

PRECISION @ 5

Relevant in Top 5:	**3**
Calculation:	3 / 5
Result:	**0.60**

"60% of the first page is useful."

PRECISION @ 10

Relevant in Top 10:	**4**
Calculation:	4 / 10
Result:	**0.40**

"Precision dropped as we went deeper."

R-Precision

Precision at rank **R**, where **R** is the total number of relevant documents for the current query.

$$\text{R-Prec} = r / R$$

(Relevant in top R) / (Total Relevant)

The Break-Even Point:
At this specific rank, Precision is mathematically equal to Recall.

Adaptive:
Unlike P@10, it adjusts to the query. If there are only 3 answers, it measures P@3.

Scenario: Total Relevant (R) = 5

1	Relevant	✓
2	Non-Rel	✕
3	Relevant	✓
4	Relevant	✓
5	Non-Rel	✕
6	Relevant	
7	Non-Rel	

Rank R = 5

Score = 3 / 5 = 0.60

Mean Average Precision (MAP)

THE GOLD STANDARD

The most widely used metric for evaluating ranked retrieval systems with binary relevance judgments.

Average Precision (AP)

Calculated for a **single query**. It is the average of Precision values at *every rank* where a **relevant document** is retrieved.

Mean (MAP)

Calculated for a **set of queries**. It is simply the arithmetic mean of the AP scores across all test queries.

ADVANTAGE

Rank-Sensitive: Heavily penalizes errors at the top. A relevant doc at #1 is worth much more than at #10.

LIMITATION

Hard to Interpret: What does "MAP = 0.4" mean? It doesn't map directly to a user experience (unlike "3 out of 5 results")

Calculating Average Precision

RANKED RESULTS

1. Relevant
2. Non-Relevant
3. Relevant
4. Non-Relevant
5. Relevant

PRECISION AT RELEVANT RANKS

P@1	1 / 1	= 1.00
Rank 2	Not Relevant	-
P@3	2 / 3	= 0.67
Rank 4	Not Relevant	-
P@5	3 / 5	= 0.60

AVERAGE PRECISION (AP)

$$(1.00 + 0.67 + 0.60) / 3$$

$$= 0.76$$

Calculating MAP

MAP is simply the **arithmetic mean** of Average Precision (AP) values over a set of queries.

QUERY 1

1. Rel
2. Non
3. Rel
4. Non

AP = **0.62**

QUERY 2

1. Non
2. Rel
3. Non
4. Rel

AP = **0.44**

QUERY 3

1. Rel
2. Rel
3. Rel
4. Non

AP = **1.00**

$$\text{MAP} = (0.62 + 0.44 + 1.00) / 3$$

0.69

Mean Reciprocal Rank (MRR)

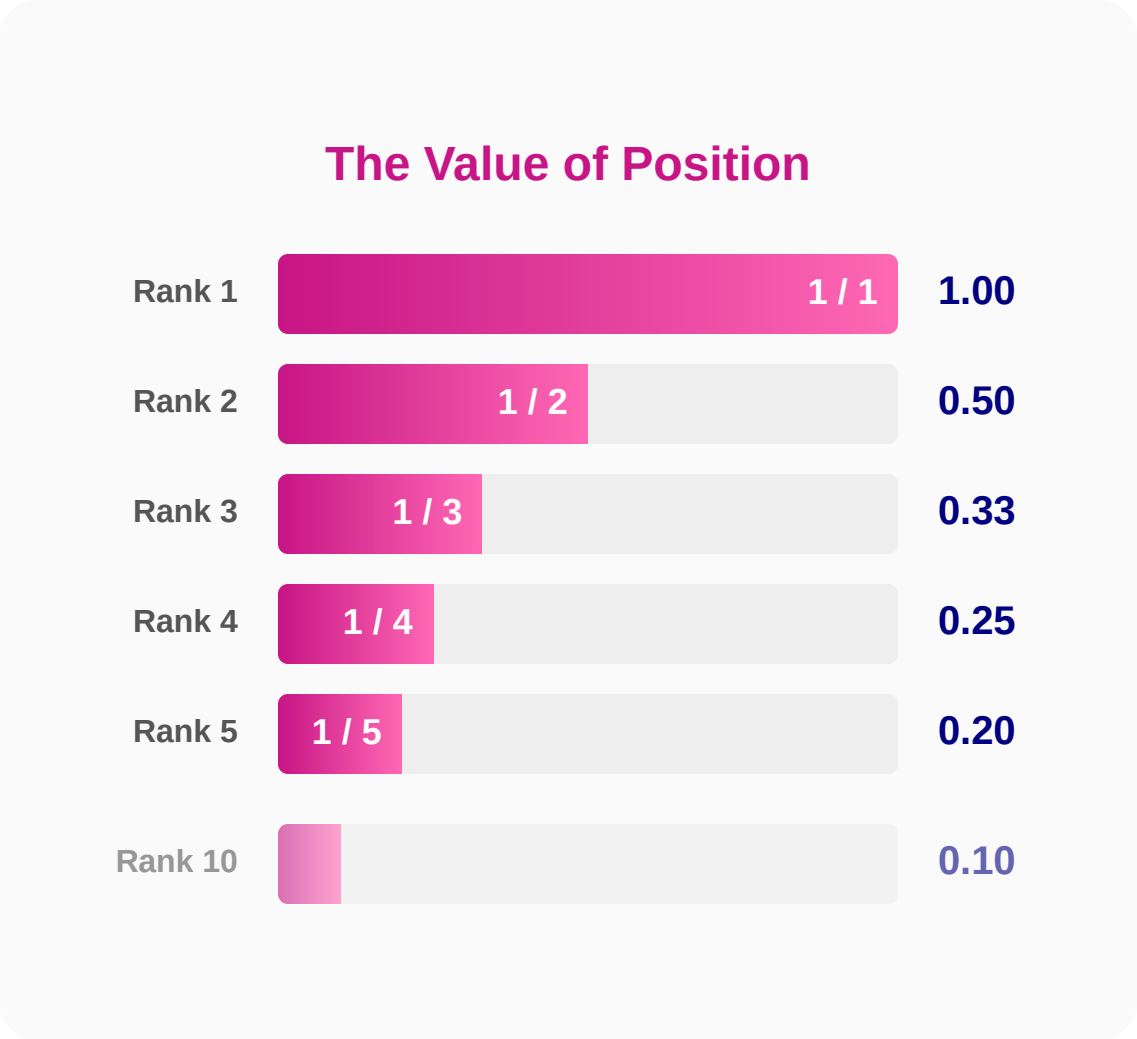
BEST FOR: NAVIGATIONAL QUERIES

When there is only **one correct answer** (e.g., "youtube", "capital of France"). We want it at Rank 1.

$$MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}$$

*Average of the reciprocal ranks of the **first** relevant document for each query.*

Harsh Penalty: Being at Rank 2 is "half as good" as Rank 1. Being at Rank 10 is nearly worthless (0.1).



MRR Calculation Example

QUERY	RESULTS (TOP 5)	RANK (K)	RECIPROCAL ($1/K$)
"cat food"	**1** 2 3 4 5	1	1.00
"best laptop"	1 2 **3** 4 5	3	0.33
"python tutorial"	1 **2** 3 4 5	2	0.50

SUM OF RR

1.83

÷

NUM QUERIES

3

=

MEAN RECIPROCAL RANK

0.61

Beyond Binary: Graded Relevance

BINARY VIEW

Rel VS Non

Traditional metrics like Precision and Recall assume a document is either **100% useful** or **0% useful**. This ignores the nuance of "better" answers.

GRADED SCALE

4 — **Perfect / Vital**
The exact answer. User needs nothing else.

3 — **Excellent**
High quality, comprehensive, very useful.

2 — **Good**
Relevant, but may need other docs to complete task.

1 — **Fair**
Mentions the topic, but marginally useful.

0 — **Bad / Spam**
Irrelevant, off-topic, or low quality.

Modern metrics like **NDCG** are designed to handle these fine-grained judgments.

Cumulative Gain (CG)

DEFINITION

The sum of the graded relevance scores of all retrieved documents up to a specific rank position p.

$$CG_p = \sum_{i=1}^{p} rel_i$$

Where rel_i is the graded relevance score at rank i.

Major Flaw: CG does not care about *order*. Swapping a highly relevant document from Rank 1 to Rank 100 does not change the CG score.

	DOCUMENT	REL SCORE	CUMULATIVE
1	Doc A (Perfect)	3	3
2	Doc B (Good)	2	5
3	Doc C (Perfect)	3	8
4	Doc D (Bad)	0	8
5	Doc E (Fair)	1	9

CG@5 = 9

Discounted Cumulative Gain (DCG)

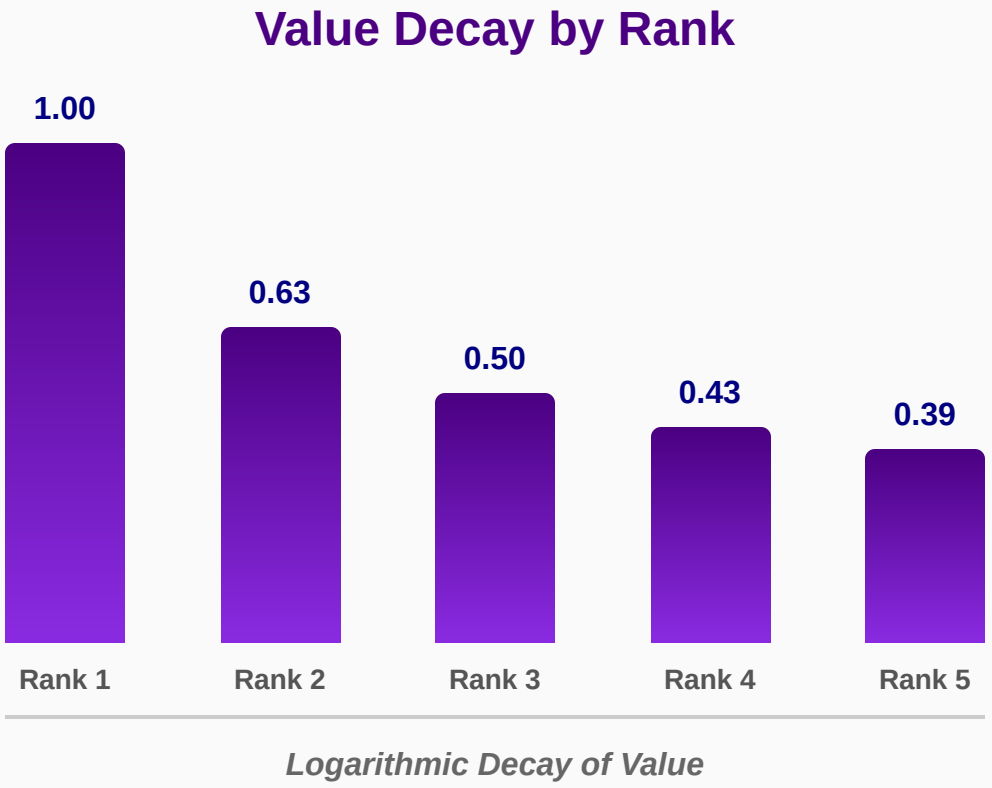
THE PROBLEM WITH CG

Cumulative Gain (CG) assumes a relevant document is equally valuable at **Rank 1** and **Rank 100**. This ignores user behavior.

THE SOLUTION: DISCOUNTING

We must **penalize** relevant documents that appear lower in the list. The value should decay as rank increases.

$$\text{Discount Factor} = \frac{1}{\log_2(\text{rank} + 1)}$$

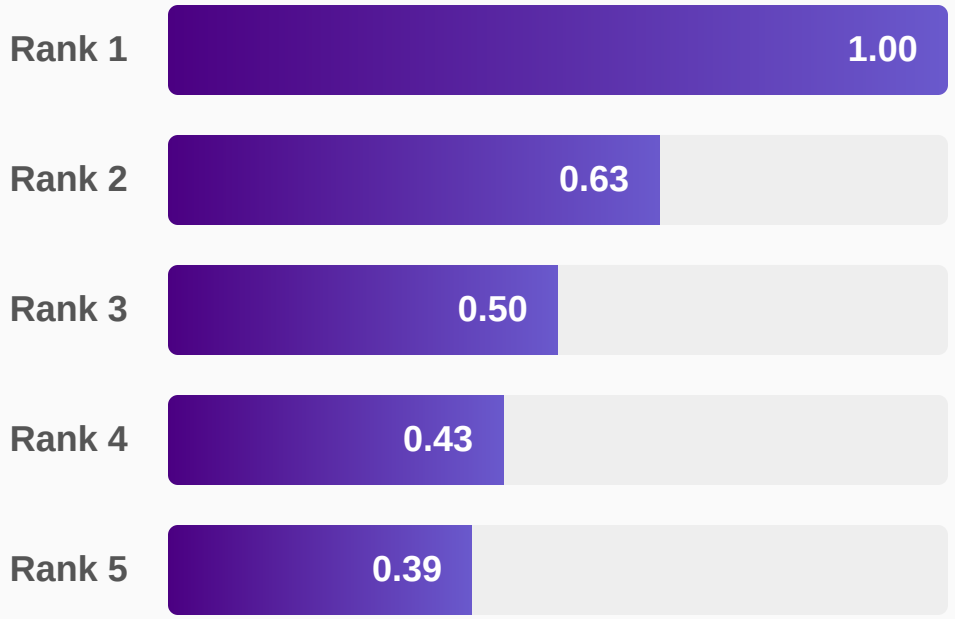


DCG Formula

$$DCG_p = \sum_{i=1}^{p} \frac{rel_i}{\log_2(i + 1)}$$

- rel_i — Graded relevance score at rank *i*
- i — The rank position (1, 2, 3...)
- $\log$ — Logarithmic discount factor

The Discount Factor: $1 / \log_2(i+1)$



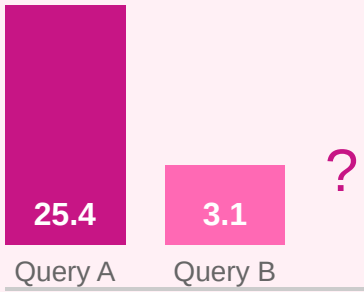
*Insight: The value of a document drops by **50%** by Rank 3.*

Normalized DCG (NDCG)

The Problem

DCG scores are not comparable across queries.

Query A might have 50 relevant results (High DCG). Query B might only have 2 relevant results (Low DCG).



Is Query A handled better? Not necessarily. It just had more opportunity.

The Solution

Normalize by the "Ideal" Score

We calculate the **Ideal DCG (IDCG)**: the score we *would* get if the ranking were perfect.

$$NDCG = \frac{DCG}{IDCG}$$

Result is always between **0.0** and **1.0**.

IDCG Calculation: Take the actual relevance scores of the retrieved documents, sort them from highest to lowest, and calculate DCG.

NDCG Step 1: Calculating DCG

Discounted Cumulative Gain (DCG) measures the total gain of a document list, discounting the gain of documents lower in the ranking.

$$DCG_p = rel_1 + \Sigma (rel_i / \log_2(i))$$

rel_i: Graded relevance score at rank *i*.
$\log_2(i)$: Discount factor. Penalizes lower ranks. Rank 1 is not discounted ($\log_2(1) = 0$, so we handle it separately).

Example: Top 5 results with relevance scores:

`[3, 2, 3, 0, 1]`

Rank (i)	Rel (rel_i)	$\log_2(i)$	Discounted (rel_i / $\log_2 i$)	DCG
1	3	-	3.00	3.00
2	2	1.00	2.00	5.00
3	3	1.58	1.89	6.89
4	0	2.00	0.00	6.89
5	1	2.32	0.43	**7.32**

Final DCG_5 = 7.32

NDCG Step 2: Ideal DCG (IDCG)

Calculating the maximum possible score for normalization

Definition: IDCG (Ideal Discounted Cumulative Gain) is the DCG score of the **perfect ranking**.

To calculate IDCG, we must reorder the retrieved documents by their relevance scores in descending order.

The Sorting Step

Original

[3, 2, 3, 0, 1, 2]

→

Ideal (Sorted)

[3, 3, 2, 2, 1, 0]

Why? We need a baseline to compare our actual ranking against. The "Ideal" ranking represents the best possible performance a system could achieve for this query.

IDCG Calculation Example

Rank (i)	Rel (Sorted)	$\log_2(i+1)$	Gain / Discount	Cumulative (IDCG)
1	**3**	1.000	3.000	3.000
2	**3**	1.585	1.893	4.893
3	**2**	2.000	1.000	5.893
4	**2**	2.322	0.861	6.754
5	**1**	2.585	0.387	7.141
6	**0**	2.807	0.000	**7.141**

$IDCG_6$ = 7.141

(This is the max possible score)

NDCG Step 3: Normalize by Ideal DCG

Calculating the final Normalized Discounted Cumulative Gain

The Formula

$$NDCG_p = DCG_p / IDCG_p$$

Why Normalize?

- **Fair Comparison:** Queries have different numbers of relevant documents.
- **Scale Independence:** A query with 50 relevant docs naturally has a higher DCG than one with only 2.
- **Standardization:** Converts the score to a **0 to 1** range.

Example Calculation

Actual DCG (from Step 1):	**4.26**
Ideal DCG (from Step 2):	**4.54**

Final Calculation:

4.26 ÷ 4.54

= 0.938

"This ranking is 93.8% as good as the perfect ranking."

Benchmark Campaigns

TREC (US)

Text REtrieval Conference

- Organized by NIST (National Institute of Standards and Technology).
- Started in 1992; the "gold standard" for IR evaluation.
- Tracks: Web, QA, Legal, Medical, Deep Learning.
- Provides large-scale test collections (documents + queries + judgments).

CLEF (Europe)

Conference and Labs of the Evaluation Forum

- Focus on **multilingual** and **cross-lingual** information access.
- Started in 2000 as a track of TREC, now independent.
- Tracks: eHealth, ImageCLEF, LifeCLEF (biodiversity), Pan (plagiarism).
- Crucial for European language evaluation.

NTCIR (Asia)

NII Testbeds and Community for Information access Research

- Based in Japan (NII). Focus on **East Asian languages**.
- Languages: Chinese, Japanese, Korean (CJK), and English.
- Tracks: Math IR, Medical NLP, We Search, QA Lab.
- Addresses specific challenges of Asian character sets.

FIRE (South Asia)

Forum for Information Retrieval Evaluation

- Based in India. Focus on **South Asian languages**.
- Languages: Hindi, Bengali, Tamil, Marathi, etc.
- Tracks: Mixed-script IR, Hate Speech Detection, Legal IR.
- Crucial for low-resource language evaluation in the region.

TREC (Text REtrieval Conference)

History & Mission

- **Founded in 1992**: Co-sponsored by the National Institute of Standards and Technology (NIST) and DARPA.
- **Goal**: To encourage research in information retrieval from large-scale text collections.
- **Standardization**: Provides a common ground for comparing retrieval algorithms using shared data and queries.

The "Cranfield at Scale"

TREC scaled up the Cranfield paradigm from thousands of documents to millions (and eventually billions in the Web track).

Major Research Tracks

Ad-hoc Track

The "standard" retrieval task. System searches a static set of documents for new queries. Investigates core ranking algorithms.

Web Track

Focuses on retrieval from large web crawls (e.g., ClueWeb). Deals with scale, link analysis, and spam.

QA Track (Question Answering)

Systems must return exact answers to natural language questions, not just document lists.

Legal & Medical Tracks

Domain-specific retrieval. High recall requirements for eDiscovery (Legal) and clinical decision support (Medical).

User Utility Evaluation

> *"Relevance is necessary, but not sufficient for user satisfaction."*

1 System Performance

Latency matters. Users perceive slow results as less relevant.

Example: Amazon found 100ms delay = 1% sales drop.

2 User Interface (UI)

Presentation quality. Are snippets informative? Is the layout readable? Does it support scanning (F-pattern)?

3 User Effort

Interaction cost. How many queries did it take? How many clicks? Did the user have to reformulate?

4 Task Success

Outcome. Did the user actually find what they needed (informational) or complete the transaction (navigational)?

Measured via: **A/B Testing** (Online) and **User Studies** (Offline).

System Performance Metrics

Evaluation isn't just about relevance (effectiveness); it's also about **efficiency**.

1. Indexing

Indexing Speed

Number of documents processed per second (docs/sec) or GB/hour.

Update Latency

Time from document arrival to searchability (Real-time requirement).

Scalability

Performance degradation as collection size grows.

2. Querying

Query Latency

Time to return results (milliseconds). Critical for user experience (< 200ms).

Throughput

Queries Per Second (QPS) the system can handle under load.

Concurrency

Number of simultaneous users supported.

3. Storage

Index Size

Total disk space required. Usually expressed as % of raw text size.

Compression Ratio

Efficiency of compression algorithms (e.g., VByte, Gamma).

Memory Footprint

RAM required for caching and dictionary storage.

User Interaction Metrics

Measuring user behavior to infer satisfaction beyond simple clicks.

Time to First Click

The time elapsed between the query submission and the user's first click on a result.

Short time *usually implies a relevant result was found quickly (good snippet).*

Dwell Time

The duration a user spends on a clicked document before returning to the search results.

Long dwell time (>30s) *suggests satisfaction.* ***Short dwell time*** *suggests "pogo-sticking" (dissatisfaction).*

Reformulation Rate

The percentage of queries that are immediately followed by a modified query (e.g., adding terms, correcting spelling).

High rate *indicates the initial results were poor or the user is struggling to express their*

A/B Testing (Online Evaluation)

Evaluating system performance by running experiments on **live user traffic**.

Incoming User Traffic (100%)

▼

Random Assignment (Bucketing)

Group A (Control)

Users see the **Current System**

Baseline Metric

Group B (Treatment)

Users see the **New Algorithm**

Experimental Metric

Why use it?

- **Realism:** Measures actual user satisfaction, not just theoretical relevance.
- **Scale:** Can test on millions of queries cheaply (once infrastructure is built).

Designing an A/B Test

1

Bucketing (Traffic Split)

How do we divide users?

- **Randomization:** Essential to remove bias.
- **User ID Hashing:** Ensure a specific user *always* sees the same version (consistency).
- **Traffic Size:** e.g., 1% to Experiment, 1% to Control, 98% to Production.

2

Execution (A vs. B)

Running the experiment.

- **Control (A):** The current system (Baseline).
- **Treatment (B):** The new algorithm or feature.
- **Isolation:** Ensure ONLY the variable being tested is different.
- **Duration:** Run long enough (e.g., 1-2 weeks) to capture weekly cycles.

3

Analysis

Making the decision.

- **Compare Metrics:** CTR, Conversion Rate, Revenue.
- **Statistical Significance:** Is the difference real or just noise? (p-value < 0.05).
- **Guardrails:** Ensure latency didn't increase, even if relevance improved.

Interleaving

A highly sensitive method for comparing two ranking functions (A and B) by **mixing their results** into a single list shown to users.

Ranking A

- Doc 1 (A)
- Doc 2 (A)
- Doc 3 (A)
- Doc 4 (A)

Interleaved List

1. Doc 1 (from A)
2. Doc 5 (from B)
3. Doc 2 (from A)
4. Doc 6 (from B)
5. Doc 3 (from A)

Ranking B

- Doc 5 (B)
- Doc 1 (B) (Duplicate)
- Doc 6 (B)
- Doc 7 (B)

(Team Draft Method: Pick top from A, then top from

Click Models

The Problem: Position Bias

Clicks ≠ Relevance.

Users are much more likely to click on the top result, regardless of its quality.

Examination Hypothesis

A user clicks a document (C) only if they **examine** it (E) AND find it **relevant** (R).

$$P(C) = P(C|E) \times P(E)$$

P(E) drops drastically with rank. At Rank 1, P(E) ≈ 1.0. At Rank 10, P(E) ≈ 0.05.

The Cascade Model

Statistical Significance

The Problem: Is it real?

System A MAP: **0.40** vs System B MAP: **0.41**

System B looks better, but is this improvement "real" or just random noise from the specific set of queries we chose?

1. The Null Hypothesis (H_0)

We start by assuming there is **no difference** between the systems.

Any observed difference is just due to chance (sampling error).

2. The p-value

The probability of seeing this data *if* the Null Hypothesis were true.

Low p-value (< 0.05) = Highly unlikely to be chance. We **reject** H_0 and call the result "Significant".

The Solution: Paired t-test

Significance Test Example (Paired t-test)

Per-Query Performance (AP)

Query	Sys A	Sys B	Diff (d)
Q1	0.20	0.40	**+0.20**
Q2	0.25	0.25	**0.00**
Q3	0.10	0.30	**+0.20**
Q4	0.50	0.45	**-0.05**
Q5	0.30	0.45	**+0.15**
		Mean ($\bar{d}$):	**0.10**

The Calculation

1. Mean Difference ($\bar{d}$): 0.10

2. Std Dev (σ_d): ≈ 0.11

3. Standard Error: $\frac{0.11}{\sqrt{5}} \approx 0.05$

4. t-statistic:

```
t = 0.10 / 0.05 = 2.0
```

5. Critical Value (N=5, 95%):

```
t_crit = 2.776
```

Result: 2.0 < 2.776

Not Significant ($p > 0.05$)

(We need more queries to prove this improvement is real!)

Lab: Metrics in Python

1. sklearn.metrics

Best for **binary classification** tasks.

Calculates Precision, Recall, F1, Accuracy.

Input: Flat arrays (y_true, y_pred)

2. pytrec_eval

The **gold standard** for IR evaluation.

Wraps the official C++ `trec_eval` tool.

Calculates MAP, NDCG, P@k.

Input: Dictionaries (qrels, runs)

```python
import pytrec_eval
import json

# 1. Define Qrels (Ground Truth)
qrel = {
  'q1': {'d1': 1, 'd2': 0, 'd3': 1}
}

# 2. Define Run (System Output)
run = {
  'q1': {'d1': 1.0, 'd2': 0.5, 'd3': 1.5}
}

# 3. Initialize Evaluator
evaluator = pytrec_eval.RelevanceEvaluator(
  qrel, {'map', 'ndcg'}
)
```

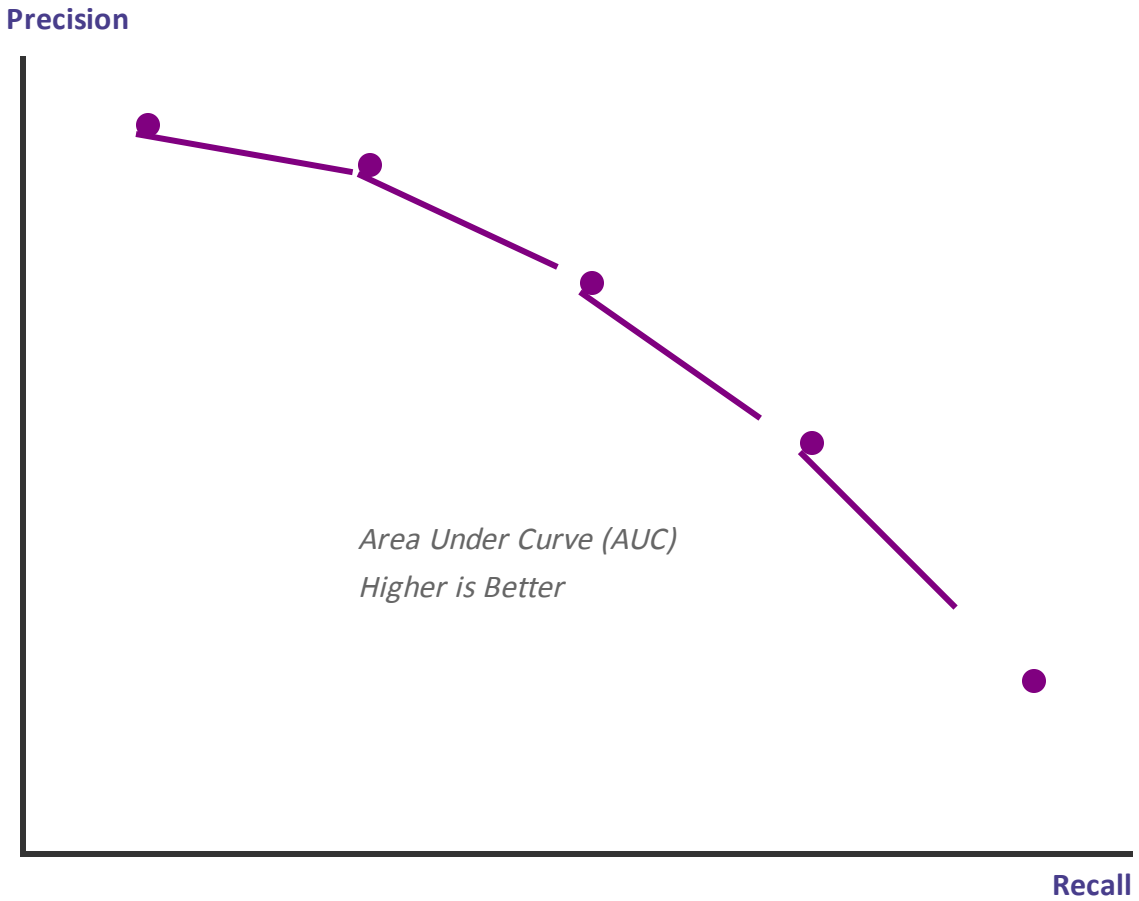
Lab: Precision/Recall Curve

Python Implementation

```
import matplotlib.pyplot as plt from
sklearn.metrics import precision_recall_curve
# 1. Get Ground Truth & Scores y_true = [0, 1,
0, 1, 1] y_scores = [0.1, 0.4, 0.35, 0.8, 0.2]
# 2. Calculate Curve Points precision, recall,
_ = precision_recall_curve(y_true, y_scores) #
3. Plot plt.plot(recall, precision,
color='purple') plt.xlabel('Recall')
plt.ylabel('Precision') plt.title('P-R Curve')
plt.show()
```

Expected Output

The curve visualizes the trade-off. A perfect system hugs the **top-right** corner (High Precision, High Recall).



Case Study: Google Search Evaluation

Google runs over **10,000 experiments** annually, but only ~1% launch. The process combines rigorous human judgment with large-scale data.

1. Human Raters (Offline)

- **Search Quality Raters:** Thousands of external contractors.
- **Guidelines:** Follow a 160+ page "Quality Rater Guidelines" document.
- **Metrics:**

  Needs Met (Fully to Fails)
  Page Quality (E-E-A-T)

"Side-by-Side" Tests

→

2. Live Traffic (Online)

- **A/B Testing:** A small % of users (e.g., 0.1%) see the new change.
- **Metrics:**

  Long Clicks (Good)
  Short Clicks (Bad)
  Time to Result

- **Safety:** Checks for latency spikes or bad results.

Real User Behavior

→

3. Launch Review

- **Launch Committee:** Senior engineers and analysts review the data.
- **Decision:** Must show statistically significant improvement in key metrics without regression in others.

Final Decision

Case Study: Netflix

The shift from **Rating Prediction** to **Ranking Optimization**.

Rating Prediction (RMSE)

Goal: Predict the exact star rating a user would give (e.g., 3.4 vs 3.8).

Metric: Root Mean Square Error.

Problem:
Users don't care about the difference between 4.1 and 4.2. They just want to know "What should I watch?"

VS

Ranking (NDCG / MAP)

Goal: Put the best items at the very top of the list.

Metric: NDCG@k, Precision@k.

Advantage:
Optimizes for **consumption**. Even if the predicted rating is wrong, as long as the order is right, the user is happy.

The Netflix Prize Lesson

"We found that small improvements in RMSE did not translate into better user experience. We shifted focus to **Top-N Ranking** and

Critique of Cranfield Paradigm

The standard model is excellent for **repeatable benchmarking**, but it abstracts away critical real-world factors.

1. Static vs. Dynamic

Assumption:

"The document collection and information needs are fixed."

Reality:
The Web changes every second. New documents appear, old ones vanish. Queries evolve (e.g., "breaking news").

2. Independence

Assumption:

"Each query is an isolated event."

Reality:
Users engage in sessions. They refine queries based on previous results. Context matters.

3. Topicality vs. Utility

Assumption:

"Relevance is objective and binary."

Reality:
Novelty matters. A relevant document is useless if the user has already read it. Diversity matters.

The "User Utility" Gap

Cranfield ignores the **interface**. A system with slightly worse ranking but a **faster UI** or better snippets might satisfy users more

User Studies (Interactive Evaluation)

Going beyond metrics to understand the **"Why"**. Observing users interacting with the system in a controlled environment.

1. Lab Studies

Users perform specific, pre-defined tasks (e.g., "Find a hotel in Paris under $200").

Measure: Success rate, time to completion.

2. Think-Aloud

Users speak their thoughts continuously while searching.

Insight: "I'm confused by this filter..." or "I didn't see that button."

3. Interviews

Post-task questionnaires or semi-structured interviews.

Goal: Understand satisfaction, trust, and frustration levels.

Advantages	Disadvantages
• Deep qualitative insights. • Identifies specific UI/UX problems. • Understands user intent.	• Expensive and time-consuming. • Small sample size (N=10-20). • **Hawthorne Effect:** Users behave differently when watched.

Eye Tracking: Gaze Plots

Recording exactly where a user looks to understand their **scanning strategy**.

1. Fixation

The eye pauses to take in information.

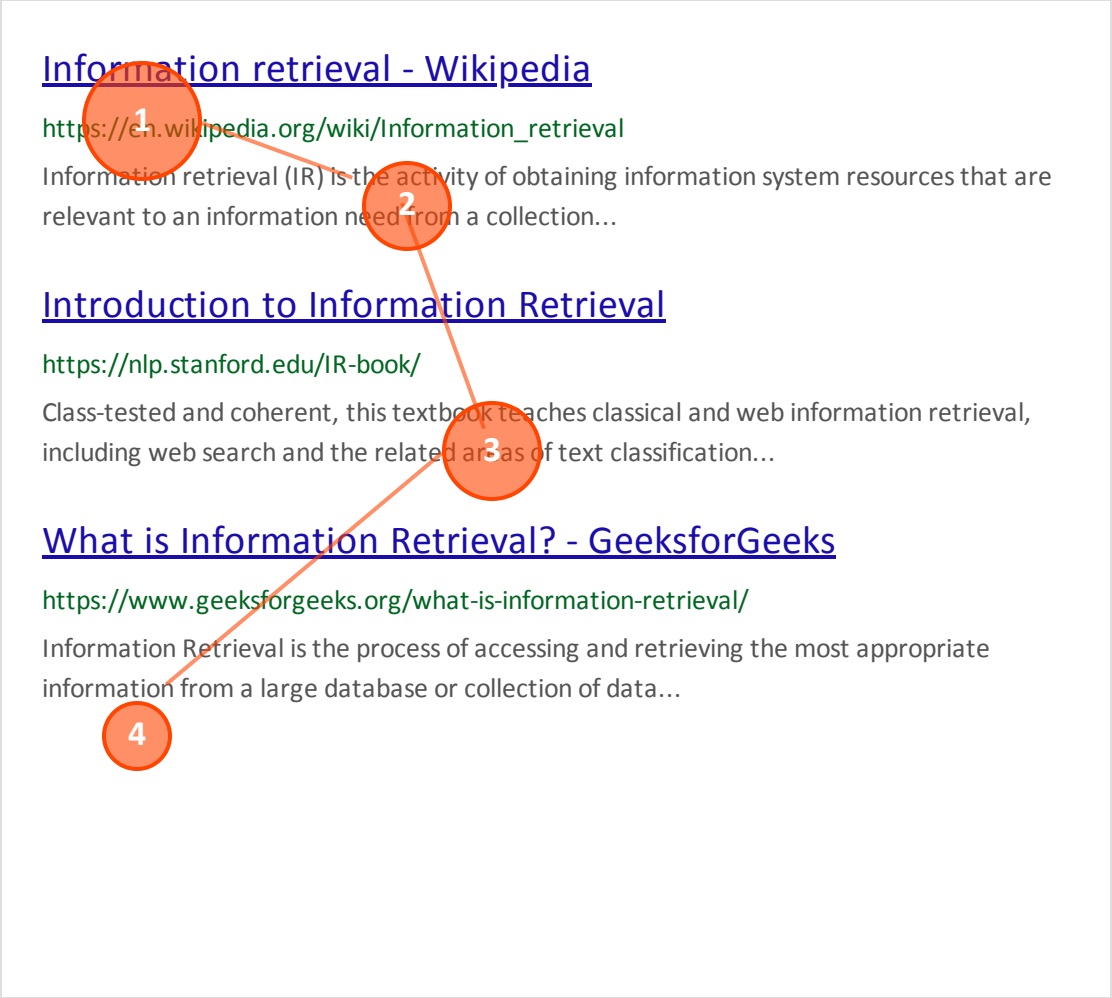
Duration: ~200-300ms

2. Saccade

Rapid movement between fixations. No visual processing happens here.

Duration: ~20-40ms

Key Insight: Users rarely read linearly. They "scan" for keywords (information scent).



The F-Shaped Pattern

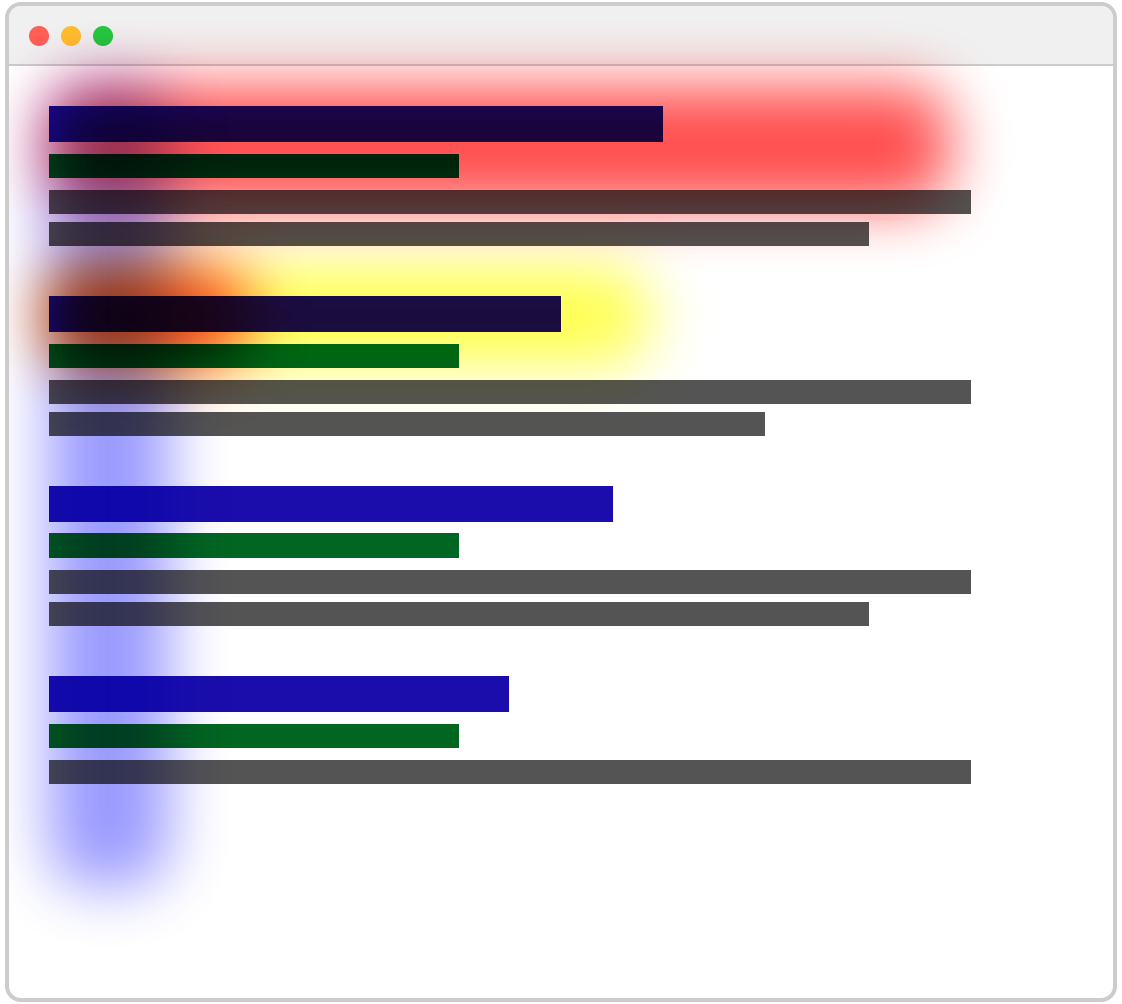
How Users Scan

Research by **Nielsen Norman Group** (Eye Tracking) reveals a dominant reading pattern for text-heavy content:

1. **Horizontal Movement 1:** Users read the top line fully.
2. **Horizontal Movement 2:** Users scan a second, shorter line.
3. **Vertical Movement:** Users scan down the left side looking for keywords.

Implication for IR

If your relevant snippet isn't in the **first two lines** or the **first few words**, it's invisible.



The Golden Triangle

information retrieval evaluation

Evaluation in Information Retrieval - Stanford NLP
https://nlp.stanford.edu/IR-book/html/htmledition/evaluation...
Standard test collections. To evaluate an information retrieval system, we need a test collection consisting of three things: A document collection...

Information Retrieval Evaluation Metrics - GeeksforGeeks
https://www.geeksforgeeks.org/information-retrieval-evaluation...
Precision and Recall are the two most common metrics used to evaluate IR systems. Precision is the fraction of retrieved documents that are relevant...

Measuring Search Effectiveness - ACM Digital Library
https://dl.acm.org/doi/10.1145/12345.67890
This paper discusses the history of evaluation in IR, starting from the Cranfield experiments to modern A/B testing methodologies...

Introduction to IR Evaluation - Coursera
https://www.coursera.org/lecture/ir-evaluation
Learn the basics of how search engines are graded and improved over time using

The "F-Pattern" of Attention

100% Visibility

The top result captures virtually all user attention. Users scan the full title and snippet.

The Drop-off

As users move down, they scan less horizontally. By Result 4, they are only scanning the first few words of the title.

"If you aren't in the triangle, you are effectively invisible."

Summary: Measuring Success

We evaluate IR systems at three levels of abstraction.

1. Binary (Sets)

"Did we find it?"

- **Precision:** Fraction of retrieved docs that are relevant. (Quality)
- **Recall:** Fraction of relevant docs that are retrieved. (Quantity)
- **F-Measure:** Harmonic mean balancing the two.

2. Ranked (Lists)

"Is it at the top?"

- **MAP:** Stability across queries. Good for general tuning.
- **MRR:** Focus on the *first* correct answer. (QA, Navigational)
- **NDCG:** Handles graded relevance. The industry standard.

3. User (Interaction)

"Is the user happy?"

- **A/B Testing:** The ultimate reality check.
- **Click Models:** Inferring relevance from behavior.
- **Utility:** Speed, UI, and novelty matter as much as relevance.

Final Thought: Optimization is dangerous without the right metric. Optimizing for Recall hurts Precision. Optimizing for Clicks can hurt Satisfaction (clickbait). **Choose your metric wisely.**

Further Reading & Resources

Core Textbook

Introduction to Information Retrieval

Manning, Raghavan, and Schütze (2008)

Chapter 8: Evaluation in Information Retrieval.
(Covers Precision/Recall, MAP, NDCG, and Kappa statistics).

Standard Tools

trec_eval

The standard C software for evaluating search engine results using TREC qrels.
https://trec.nist.gov/trec_eval/

Seminal Research Papers

- **The Significance of the Cranfield Tests**
  Cleverdon (1991)
  The foundational paper establishing the paradigm of fixed collections and relevance judgments.
- **Cumulated Gain-based Evaluation of IR Techniques**
  Järvelin & Kekäläinen (2002)
  The paper that introduced **NDCG**, arguing that binary relevance is insufficient.
- **The Philosophy of Information Retrieval Evaluation**
  Voorhees (2002)
  Discusses the stability of metrics and how much data is needed to trust a conclusion.

Review Questions

Test your understanding of evaluation metrics and methodologies.

Q1

Calculation: Precision & AP

*Given a ranked list of 5 docs: **[Relevant, Non-Rel, Relevant, Non-Rel, Non-Rel]**.*

*Calculate **Precision@3** and **Average Precision (AP)**.*

Q2

Concept: Pooling

Explain the "Pooling" method used in TREC. Why is it necessary for large collections? Does it bias the results?

Q3

Metric Selection

*For a **navigational query** (e.g., "facebook login"), which metric is most appropriate: MAP, NDCG, or MRR? Why?*

Q4

User Behavior

*How does **position bias** affect the analysis of click-through logs? How can we correct for it?*